

## EXPERIMENT 2

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### Question A:

(A)

(i) You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

(B)

(ii) Make all the rankings in descending order except **India** and **Sri Lanka**.

| S.No | Rankings | Country     |
|------|----------|-------------|
| 1    | 1        | Finland     |
| 2    | 2        | Denmark     |
| 3    | 3        | Iceland     |
| 4    | 4        | Israel      |
| 5    | 5        | Netherlands |
| 6    | 6        | Sweden      |
| 7    | 7        | Norway      |
| 8    | 126      | India       |
| 9    | 128      | Sri Lanka   |

### Solution:

#### 1. Query:

A)

```
SELECT rankings, country
FROM tbl_happiness
ORDER BY
    CASE
        WHEN country = 'India' THEN 1
        WHEN country = 'Sri Lanka' THEN 2
        ELSE 3
    END,
rankings ;
```

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B)

```
SELECT rankings, country
FROM tbl_happiness
ORDER BY
CASE
  WHEN country = 'India' THEN 1
  WHEN country = 'Sri Lanka' THEN 2
  ELSE 3
END,
rankings desc;
```

### 2. Output

|   | rankings<br>integer | country<br>character varying (50) |
|---|---------------------|-----------------------------------|
| 1 | 126                 | India                             |
| 2 | 128                 | Sri Lanka                         |
| 3 | 1                   | Finland                           |
| 4 | 2                   | Denmark                           |
| 5 | 3                   | Iceland                           |
| 6 | 4                   | Israel                            |
| 7 | 5                   | Netherlands                       |
| 8 | 6                   | Sweden                            |
| 9 | 7                   | Norway                            |

|   | rankings<br>integer | country<br>character varying (50) |
|---|---------------------|-----------------------------------|
| 1 | 126                 | India                             |
| 2 | 128                 | Sri Lanka                         |
| 3 | 7                   | Norway                            |
| 4 | 6                   | Sweden                            |
| 5 | 5                   | Netherlands                       |
| 6 | 4                   | Israel                            |
| 7 | 3                   | Iceland                           |
| 8 | 2                   | Denmark                           |
| 9 | 1                   | Finland                           |

Question B:

(C)

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

| employee_id | name    | reports_to | age |
|-------------|---------|------------|-----|
| 9           | Hercy   | NULL       | 43  |
| 6           | Alice   | 9          | 41  |
| 4           | Bob     | 9          | 36  |
| 2           | Winston | NULL       | 37  |

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Query:

```
SELECT
    E2.employee_id,
    E2.name,
    COUNT(E1.employee_id) AS reports_count,
    ROUND(AVG(E1.age),0) AS average_age
FROM Employees E1
JOIN Employees E2
ON E1.reports_to = E2.employee_id
GROUP BY E2.employee_id, E2.name
HAVING COUNT(E1.employee_id) > 1
ORDER BY E2.employee_id;
```

Output:

| Data Output Messages Notifications                                                  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |
|  |  |  | SQL                                                                                 |                                                                                     |
|                                                                                     | employee_id<br>[PK] integer                                                         | name<br>character varying (50)                                                      | reports_count<br>bigint                                                             | average_age<br>numeric                                                              |
| 1                                                                                   | 9                                                                                   | Hercy                                                                               | 2                                                                                   | 39                                                                                  |